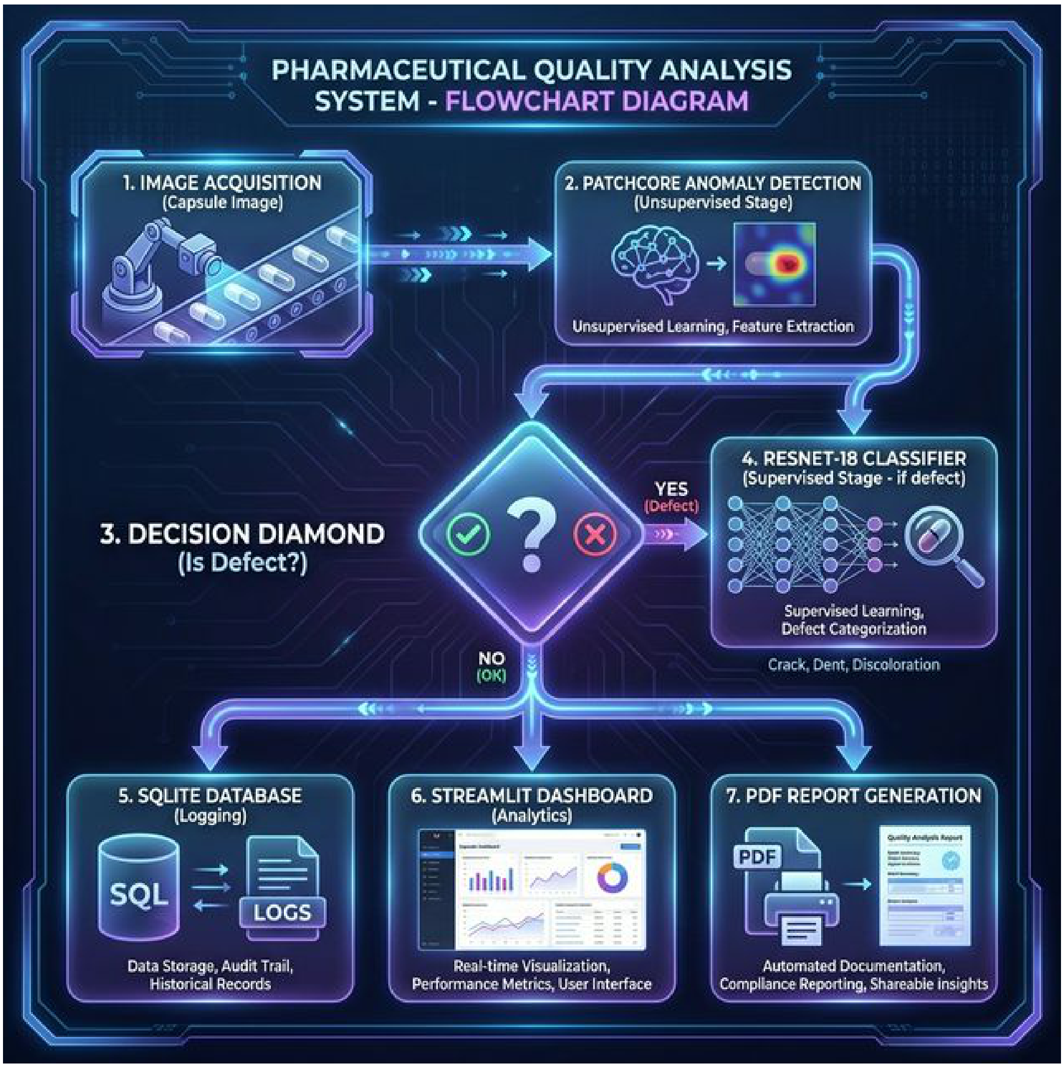
Final Year Project - Detailed Technical Report

System: Integrated Pharmaceutical Quality Analysis and Reporting

1. System Architecture

The system adopts a Hybrid Deep Learning Architecture designed for high-precision defect detection and root-cause analysis. It integrates unsupervised anomaly detection with supervised classification.



Database Schema: The SQLite3 database (capsule_inspections.db) stores inspection IDs, status (Good/Defect), defect type, anomaly score, severity, and timestamps.

2. Dataset Details

Source: MVTec AD Capsule Dataset.

Size: 219 Good images for training, 132 images for testing.

Features: 256x256 RGB images.

Preprocessing: Resizing, normalization (ImageNet stats), and data augmentation for the classifier (rotations, flips).

3. Algorithms / Models Implemented

Stage 1: PatchCore (Unsupervised Anomaly Detection). Extracts locally aware features from WideResNet-50 and uses coreset sampling for efficient memory bank storage.

Stage 2: ResNet-18 (Supervised Classification). Fine-tuned to categorize defects like Cracks, Pokes, and Scratches.

4. Implementation Status

The system is in the Integration Phase. Key modules like app.py, db.py, analytics.py, and report.py are fully functional.

Code Structure: Organized into core logic (analytics, db, report) and execution scripts (app.py, train.py).

5. Testing Strategy

1. Unit Testing: Verified database persistence and PDF logic.
2. Integration Testing: Validated the full inference pipeline.
3. Test Cases: Successful detection of all major defect types in the test set.

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6. Preliminary Results

Accuracy: PatchCore achieves ~98% AUC on capsule anomalies. ResNet-18 classifier shows ~92% classification accuracy.

Speed: Inference takes <500ms on CPU.

7. Challenges & Future Enhancements

Challenges: Handling large weight files and lighting variations.

Future: Edge deployment (NVIDIA Jetson) and PLC integration.

8. Updated Gantt Chart (Deadline: March 2026)